IGCSE 0607 Practice Questions — E4.3 — Combined Questions & Markscheme

Length and midpoint

Total Questions: 18 — **Total Marks:** 121 Marks

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1 The points A and B have coordinates $(2, -3)$ and $(8, 5)$ respectively.

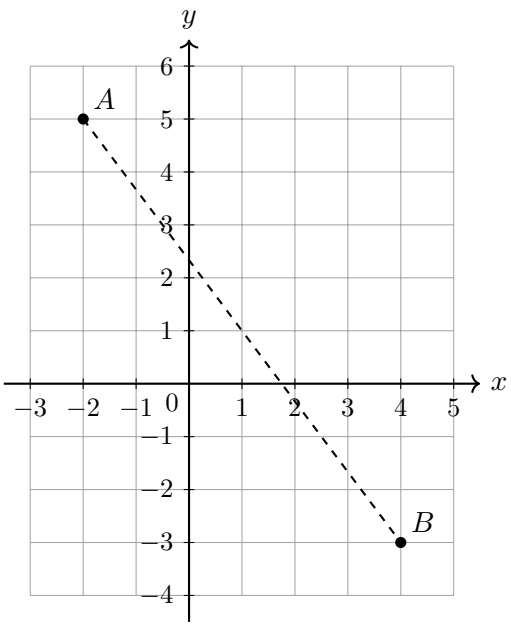
(a) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

(b) Calculate the length of the line segment AB.

$AB =$ ..... **[2]**

2 The points $A(-2, 5)$ and $B(4, -3)$ are shown on the coordinate grid.



(a) Find the coordinates of the midpoint of the line segment AB.

((..... , ...)) **[2]**

(b) Calculate the exact length of the line segment AB.

..... **[3]**

3 Point A has coordinates $(1, 4)$ and point B has coordinates $(7, -4)$.

(a) Calculate the coordinates of the midpoint of the line segment AB.

(..... , ...) **[2]**

(b) Calculate the length of the line segment AB.

..... **[3]**

4 The points A and B have coordinates $(2, -3)$ and $(8, 5)$ respectively.

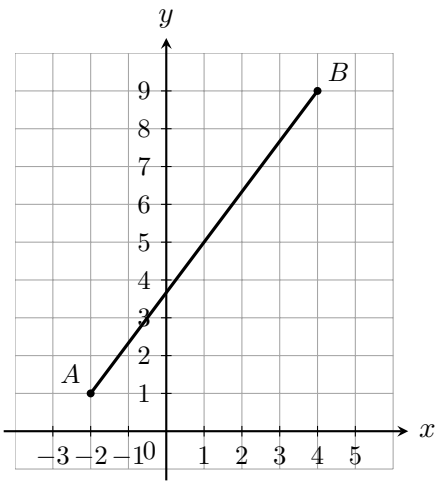
(a) Find the coordinates of the midpoint of the line segment AB.

(..... , ...) **[2]**

(b) Calculate the exact length of the line segment AB.

..... **[3]**

5 The diagram shows a line segment AB plotted on a Cartesian coordinate grid.



(a) Write down the coordinates of point A and point B.

$A =$ ..... , $B =$ ..... **[2]**

(b) Find the coordinates of the midpoint M of the line segment AB.

$M =$ (..... , ...) **[2]**

(c) Calculate the length of the line segment AB.

..... units **[2]**

6 The point $M(4, 1)$ is the midpoint of the line segment joining $P(-2, 7)$ and Q.

(a) Find the coordinates of Q.

(..... , ...) **[2]**

(b) The point R has coordinates $(k, 10)$, where k is a constant.
The length of PR is 5.
Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

7 A circle has its centre at the origin, $O(0, 0)$.
The point $P(-3, 4)$ lies on the circumference of this circle.

(a) Work out the radius of the circle.

..... **[2]**

(b) The line segment PQ is a diameter of the circle.
Find the coordinates of Q.

(..... , ...) **[2]**

(c) The point R has coordinates $(k, \sqrt{11})$.
R also lies on the circumference of the circle, and k is a negative number.
Work out the exact value of k.

$k =$ ..... **[3]**

8 The line segment PQ has midpoint $M(3, -1)$.
The coordinates of point P are $(-2, 4)$.

(a) Find the coordinates of point Q.

$Q = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[2]**

(b) Calculate the exact length of PQ, giving your answer in the form $k\sqrt{2}$, where k is an integer.

$PQ = \ldots\ldots\ldots\ldots\ldots$ **[3]**

(c) Point R lies on the y-axis such that $PR = QR$.
Find the coordinates of R.

$R = (\ldots\ldots\ldots\ ,\ \ldots\ldots\ldots)$ **[4]**

9 The points $A(-2,-1)$, $B(4,3)$ and $C(8,-3)$ form a triangle.
M is the midpoint of AB and N is the midpoint of AC.

(a) Find the coordinates of M and the coordinates of N.

$M =$ ..... , $N =$ ..... **[2]**

(b) Show that the length of MN is exactly half the length of BC.

[4]

10 Points A and B have coordinates $(3p, p-2)$ and $(p, 2p+1)$ respectively, where p is a constant.

(a) The midpoint of AB lies on the line $y = x + 1$.
Find the value of p.

$p =$ ..... **[3]**

(b) Given instead that the length of the line segment AB is 6, show that $5p^2 + 6p - 27 = 0$.

[3]

(c) Solve the equation $5p^2 + 6p - 27 = 0$ to find the two possible values of p.

$p =$ ..... , $p =$ ..... **[3]**

11 A circle has diameter AB, where A is the point $(-1, 2)$ and B is the point $(5, 8)$.

(a) Find the coordinates of the centre of the circle, C.

(..... , ...) **[2]**

(b) Find the exact radius of the circle.
Give your answer in the form $a\sqrt{b}$, where a and b are positive integers and b is as small as possible.

..... **[3]**

(c) Point D is such that the midpoint of CD is $M\left(\frac{5}{2}, \frac{3}{2}\right)$.

Find the exact length of AD, giving your answer in the form $c\sqrt{d}$, where c and d are integers and d is as small as possible.

..... **[5]**

12 The points A and B have coordinates $(-2, 5)$ and $(4, -3)$ respectively.

(a) Calculate the exact length of the line segment AB.

..... **[3]**

(b) Find the coordinates of the midpoint of AB.

(..... , ...) **[2]**

13 Point P has coordinates $(3, k)$ and point Q has coordinates $(7, 5)$.
The length of the line segment PQ is 5 units.

(a) Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

(b) M is the midpoint of PQ.
For the case where $k < 5$, find the coordinates of M.

((..... , ...)) **[2]**

14 The midpoint of the line segment PQ is $M(1, -2)$.
The coordinates of P are $(-3, 5)$.

(a) Find the coordinates of Q.

(..... , ...) **[2]**

(b) Calculate the length of the line segment PQ.
Give your answer correct to 3 significant figures.

..... **[3]**

15 Point P has coordinates $(-3, 5)$ and point Q has coordinates $(k, -1)$.
The length of the line segment PQ is 10 units.

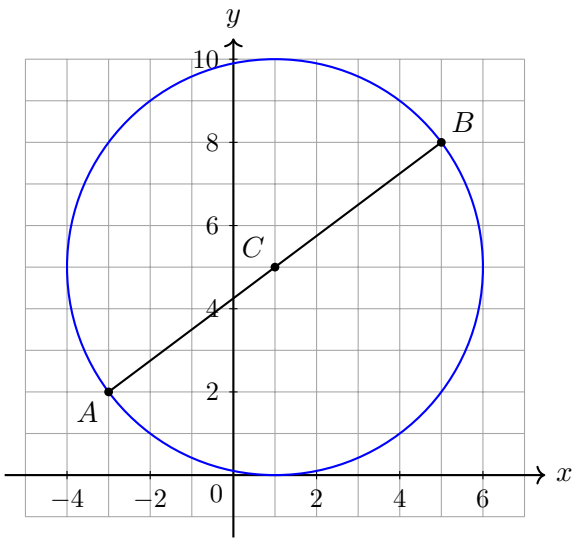
(a) Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[4]**

(b) Given that $k > 0$, point M is the midpoint of PQ.
Find the coordinates of M.

(..... , ...) **[2]**

16 The points $A(-3, 2)$ and $B(5, 8)$ are the endpoints of a diameter of a circle.



(a) Find the coordinates of the centre of the circle, C.

((..... , ...)) **[2]**

(b) Calculate the area of the circle.
Give your answer correct to 1 decimal place.

..... **[3]**

(c) The point $D(k, 8)$ lies on the circumference of the circle.
Find the possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

17 The points $A(2p, p)$ and $B(3p, 4p)$ are joined by a straight line, where $p > 0$.

(a) The length of the line segment AB is $5\sqrt{10}$.
Find the value of p.

$p =$ ..... **[4]**

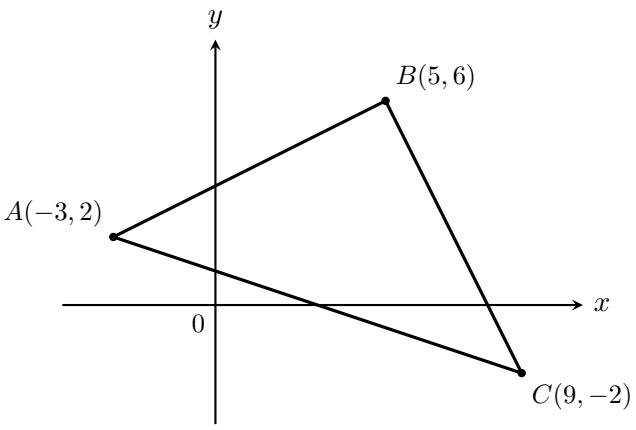
(b) Point C lies on the line segment AB such that the ratio $AC : CB = 2 : 3$.
Using your value of p from part (a), find the coordinates of C.

(..... , ...) **[3]**

(c) The point $D(x, y)$ is such that C is the midpoint of BD.
Work out the coordinates of D.

(..... , ...) **[2]**

18 The diagram shows a sketch of triangle ABC with vertices $A(-3, 2)$, $B(5, 6)$, and $C(9, -2)$.



(a) Show that triangle ABC is isosceles.

[4]

(b) M is the midpoint of AC.

(i) Find the coordinates of M.

$M = (\ldots\ldots\ ,\ \ldots\ldots)$ **[2]**

(ii) Calculate the length of BM, giving your answer correct to 3 significant figures.

$BM = \ldots\ldots\ldots\ldots\ldots$ **[2]**

(c) Calculate the area of triangle ABC.

..... square units **[2]**

(d) Point D is such that $ABCD$ is a square.
Find the coordinates of D.

$D = ($ ..... , ... $)$ **[2]**

1 **(a)** $\left(\frac{2+8}{2}, \frac{-3+5}{2}\right)$ **[M1]**

$(5, 1)$ **[A1]**

(b) $\sqrt{(8-2)^2+(5-(-3))^2}$ or $\sqrt{6^2+8^2}$ **[M1]**

10 **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** $\left(\frac{-2+4}{2}, \frac{5+(-3)}{2}\right)$ **[M1]**

$(1, 1)$ **[A1]**

(b) $(4-(-2))^2+(-3-5)^2$ or $6^2+(-8)^2$ **[M1]**

$\sqrt{36+64}$ or $\sqrt{100}$ **[M1]**

10 **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** $\left(\frac{1+7}{2}, \frac{4+(-4)}{2}\right)$ **[M1]**

$(4, 0)$ **[A1]**

(b) $(7-1)^2+(-4-4)^2$ oe **[M1]**

$\sqrt{36+64}$ or $\sqrt{100}$ **[M1]**

10 **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** Uses the midpoint formula: $\left(\frac{2+8}{2}, \frac{-3+5}{2}\right)$ **[M1]**

$(5, 1)$ **[A1]**

(b) Uses the distance formula: $\sqrt{(8-2)^2+(5-(-3))^2}$ **[M1]**

$\sqrt{6^2+8^2}$ or $\sqrt{36+64}$ **[M1]**

10 **[A1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** $A = (-2, 1)$ **[B1]**

$B = (4, 9)$ **[B1]**

(b) $\left(\frac{-2+4}{2}, \frac{1+9}{2}\right)$ **[M1]**

$(1, 5)$ **[A1]**

(c) $\sqrt{(4-(-2))^2+(9-1)^2}$ or $\sqrt{6^2+8^2}$ **[M1]**

10 **[A1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** Sets up an equation for the x or y coordinate, e.g. $\frac{-2+x}{2} = 4$ or $\frac{7+y}{2} = 1$ **[M1]**

$(10, -5)$ **[A1]**

(b) $\sqrt{(k-(-2))^2 + (10-7)^2} = 5$ or $(k+2)^2 + 3^2 = 5^2$ **[M1]**

$(k+2)^2 = 16$ or $k^2 + 4k - 12 = 0$ **[M1]**

2 and -6 **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** Calculates length OP: $\sqrt{(-3)^2 + 4^2}$ or $\sqrt{9+16}$ **[M1]**

5 **[A1]**

(b) Recognises $O(0,0)$ is the midpoint of PQ, so sets up $\frac{-3+x}{2} = 0$ and $\frac{4+y}{2} = 0$ OR uses translation from P to O **[M1]**

$(3, -4)$ **[A1]**

(c) Sets up equation for length OR: $k^2 + (\sqrt{11})^2 = 5^2$ **[M1]**

$k^2 + 11 = 25$

$k^2 = 14$ [M1]

$-\sqrt{14}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** $\frac{-2+x}{2} = 3$ or $\frac{4+y}{2} = -1$ **[M1]**

$(8, -6)$ **[A1]**

(b) $PQ = \sqrt{(8-(-2))^2 + (-6-4)^2}$ **[M1]**

$\sqrt{100+100}$ or $\sqrt{200}$ **[M1]**

$10\sqrt{2}$ **[A1]**

(c) Let $R = (0, y)$ and form $(0-(-2))^2 + (y-4)^2 = (0-8)^2 + (y-(-6))^2$ **[M1]**

$4 + y^2 - 8y + 16 = 64 + y^2 + 12y + 36$ **[M1]**

$-20y = 80$ giving $y = -4$ **[A1]**

$(0, -4)$ **[A1]**

[Back to Question] *[Total Marks: 9]*

9 **(a)** $M = (1, 1)$ **[B1]**

$N = (3, -2)$ **[B1]**

(b) Calculates length $MN = \sqrt{(3-1)^2 + (-2-1)^2}$ **[M1]**

$MN = \sqrt{4+9} = \sqrt{13}$ **[A1]**

Calculates length $BC = \sqrt{(8-4)^2 + (-3-3)^2}$ **[M1]**

$BC = \sqrt{16+36} = \sqrt{52}$ and shows $\sqrt{52} = 2\sqrt{13}$, therefore $MN = \frac{1}{2}BC$ **[A1]**

[Back to Question] *[Total Marks: 6]*

10 (a) Finds coordinates of the midpoint in terms of p: $\left(\frac{4p}{2}, \frac{3p-1}{2}\right)$ or $\left(2p, \frac{3p-1}{2}\right)$ **[M1]**

Substitutes into the equation of the line: $\frac{3p-1}{2} = 2p + 1$ **[M1]**

-3 **[A1]**

(b) Sets up the distance squared equation: $(3p-p)^2 + (p-2-(2p+1))^2 = 6^2$ **[M1]**

Simplifies to $(2p)^2 + (-p-3)^2 = 36$ **[M1]**

Expands to $4p^2 + p^2 + 6p + 9 = 36$ and rearranges to $5p^2 + 6p - 27 = 0$ with no errors seen **[A1]**

(c) Factorises to $(5p-9)(p+3) = 0$ or uses correct quadratic formula **[M2]**

Award [M1] for a partial factorisation or one error in the quadratic formula

1.8 (or $\frac{9}{5}$) and -3 **[A1]**

[Back to Question] *[Total Marks: 9]*

11 (a) $\left(\frac{-1+5}{2}, \frac{2+8}{2}\right)$ **[M1]**

$(2, 5)$ **[A1]**

(b) $(2-(-1))^2 + (5-2)^2$ or $\frac{1}{2}\sqrt{(5-(-1))^2 + (8-2)^2}$ oe **[M1]**

$\sqrt{9+9}$ or $\sqrt{18}$ **[M1]**

$3\sqrt{2}$ **[A1]**

(c) $\frac{2+x}{2} = \frac{5}{2}$ and $\frac{5+y}{2} = \frac{3}{2}$ oe **[M1]**

$x = 3$ and $y = -2$ to give $D(3, -2)$ **[A1]**

Length $AD = \sqrt{(3-(-1))^2 + (-2-2)^2}$ oe **[M1]**

$\sqrt{16+16}$ or $\sqrt{32}$ **[M1]**

$4\sqrt{2}$ **[A1]**

[Back to Question] *[Total Marks: 10]*

12 (a) Uses distance formula: $\sqrt{(4-(-2))^2 + (-3-5)^2}$ **[M1]**

Simplifies to $\sqrt{6^2 + (-8)^2}$ or $\sqrt{36+64}$ **[M1]**

10 **[A1]**

(b) Uses midpoint formula: $\left(\frac{-2+4}{2}, \frac{5+(-3)}{2}\right)$ **[M1]**

$(1, 1)$ **[A1]**

[Back to Question] *[Total Marks: 5]*

13 (a) $(7-3)^2 + (5-k)^2 = 5^2$ or equivalent **[M1]**

$16 + (5-k)^2 = 25$ or $(5-k)^2 = 9$ **[M1]**

$k = 2$ and $k = 8$ **[A1]**

(b) Identifies $P(3, 2)$ to use in midpoint calculation: $\left(\frac{3+7}{2}, \frac{2+5}{2}\right)$ **[M1]**

$(5, 3.5)$ **[A1]**

[Back to Question] *[Total Marks: 5]*

14 (a) Sets up an equation for at least one coordinate, e.g., $\frac{-3+x}{2} = 1$ or $\frac{5+y}{2} = -2$ **[M1]**

$(5, -9)$ **[A1]**

(b) Uses distance formula with $P(-3, 5)$ and their $Q(5, -9)$: $\sqrt{(5-(-3))^2 + (-9-5)^2}$ **[M1]**

$\sqrt{8^2 + (-14)^2}$ or $\sqrt{64 + 196}$ or $\sqrt{260}$ **[M1]**

16.1 **[A1]**

[Back to Question] *[Total Marks: 5]*

15 (a) $(k-(-3))^2 + (-1-5)^2 = 10^2$ oe **[M1]**

$(k+3)^2 + 36 = 100$ or $k^2 + 6k + 9 + 36 = 100$ **[M1]**

$(k+3)^2 = 64$ or $k^2 + 6k - 55 = 0$ **[M1]**

$k = 5$ or $k = -11$ **[A1]**

(b) Chooses $k = 5$ to establish coordinates of Q as $(5, -1)$ **[M1]**

$(1, 2)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

16 (a) $\left(\frac{-3+5}{2}, \frac{2+8}{2}\right)$ **[M1]**

$(1, 5)$ **[A1]**

(b) Radius $r = \sqrt{(1-(-3))^2 + (5-2)^2}$ or $\sqrt{4^2 + 3^2} = 5$ **[M1]**

Area $= \pi \times 5^2$ **[M1]**

78.5 **[A1]**

(c) Distance $CD = 5 \implies (k-1)^2 + (8-5)^2 = 5^2$ **[M1]**

$(k-1)^2 + 9 = 25 \implies (k-1)^2 = 16$ **[M1]**

$k = 5$ and $k = -3$ **[A1]**

[Back to Question] *[Total Marks: 8]*

17 **(a)** Sets up the distance formula: $\sqrt{(3p-2p)^2+(4p-p)^2} = 5\sqrt{10}$ **[M1]**

Simplifies the terms inside the root: $\sqrt{p^2+(3p)^2} = 5\sqrt{10}$ **[M1]**

Squares both sides and simplifies: $10p^2 = 250$ **[M1]**

$p = 5$ (rejects -5 as $p > 0$) **[A1]**

(b) Evaluates coordinates of $A(10, 5)$ and $B(15, 20)$ **[B1]**

Sets up a valid expression for the coordinates of C, e.g., x-coordinate $= 10 + \frac{2}{5}(15-10)$ or y-coordinate $= 5 + \frac{2}{5}(20-5)$ **[M1]**

$(12, 11)$ **[A1]**

(c) Uses the midpoint formula, e.g. $\frac{15+x}{2} = 12$ and $\frac{20+y}{2} = 11$, or $D = 2C - B$ **[M1]**

$(9, 2)$ **[A1]**

[Back to Question] *[Total Marks: 9]*

18 **(a)** $AB = \sqrt{(5-(-3))^2+(6-2)^2} = \sqrt{80}$ $(= 8.94\ldots)$ **[M1]**

$BC = \sqrt{(9-5)^2+(-2-6)^2} = \sqrt{80}$ $(= 8.94\ldots)$ **[M1]**

$AC = \sqrt{(9-(-3))^2+(-2-2)^2} = \sqrt{160}$ $(= 12.6\ldots)$ **[M1]**

$AB = BC$ (and $ACeAB$), therefore triangle ABC is isosceles **[A1]**

(b) **(i)** $\left(\frac{-3+9}{2}, \frac{2+(-2)}{2}\right)$ **[M1]**

$(3, 0)$ **[A1]**

(ii) $BM = \sqrt{(5-3)^2+(6-0)^2}$ $(= \sqrt{40})$ **[M1]**

6.32 **[A1]**

(c) $\frac{1}{2} \times \sqrt{160} \times \sqrt{40}$ or $\frac{1}{2} \times \sqrt{80} \times \sqrt{80}$ **[M1]**

40 **[A1]**

(d) $\frac{5+x}{2} = 3$ and $\frac{6+y}{2} = 0$ (or vector translation $B + \overrightarrow{AD}$) **[M1]**

$(1, -6)$ **[A1]**

[Back to Question] *[Total Marks: 12]*